Pool-ball localisation and recognition

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1 Processing and design

The system locates balls and assigns type, colour and identity. AllBalls supplies parameter selection and colour references; Tables1–6 supply repeatedly inspected validation images rather than an independent test set.

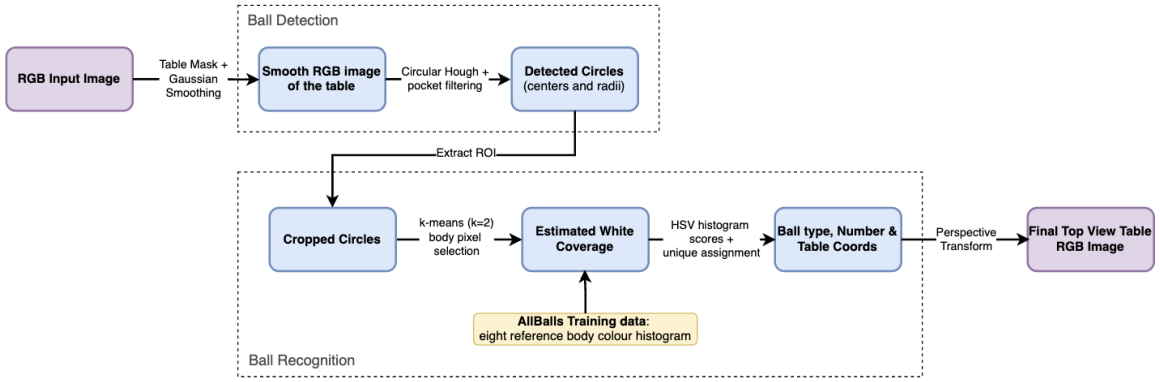


Figure 1: Final Detection and Recognition pipeline.

1.1 Table restriction and circular detection

The quadrilateral through $(337, 35)$, $(1570, 21)$, $(1888, 451)$ and $(38, 472)$ masks the background. Greyscale intensities are smoothed with a 5×5 Gaussian, $\sigma = 1.5$. Circular Hough voting [1] estimates centres and radii before perspective correction, which could distort ball outlines.

A 480-configuration search on AllBalls ranked matched balls first, then fewer false detections and lower centre error. Selected settings: accumulator resolution ratio 1.5, centre separation 18 pixels, edge threshold 75, centre-vote threshold 35, and radii 15–30 pixels. Candidates within 25 table units of six approximate pocket centres are rejected; this is an exclusion threshold, not a measured pocket radius.

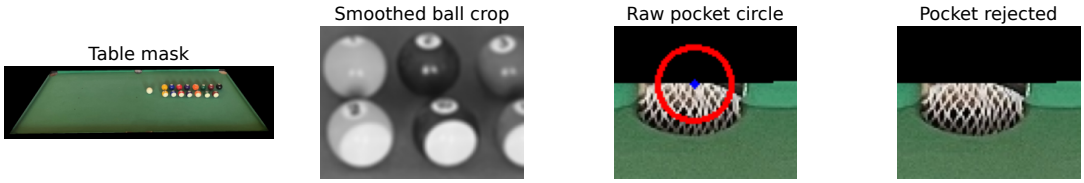


Figure 2: AllBalls: full table mask, smoothed ball detail, and the same pocket before/after filtering. The raw 17 candidates become 16 matched detections.

1.2 Ball regions, type estimation and colour references

Each retained circle defines a square colour crop and a circular pixel mask. Only masked pixels enter recognition; the crop is clipped at the image boundary. Two BGR clusters estimate the proportions of light and non-light material. The fixed clustering seed and three initialisations make the comparison reproducible; termination uses at most 50 iterations or a centre-change threshold of 0.2. Spatial position is not a clustering feature: disconnected patches may share a cluster.

A cluster's representative colour is converted to HSV. It is light when $V \geq 170$ and $S \leq 165$; otherwise $V \leq 90$ denotes dark material. Cluster proportions estimate the light fraction w. The initial classification is cue-like white for $w \geq 0.85$, solid for non-light coverage at least 70%, and striped for other mixtures with both light and non-light material. These manually selected rules estimate visible appearance, not the ball's hidden surface.

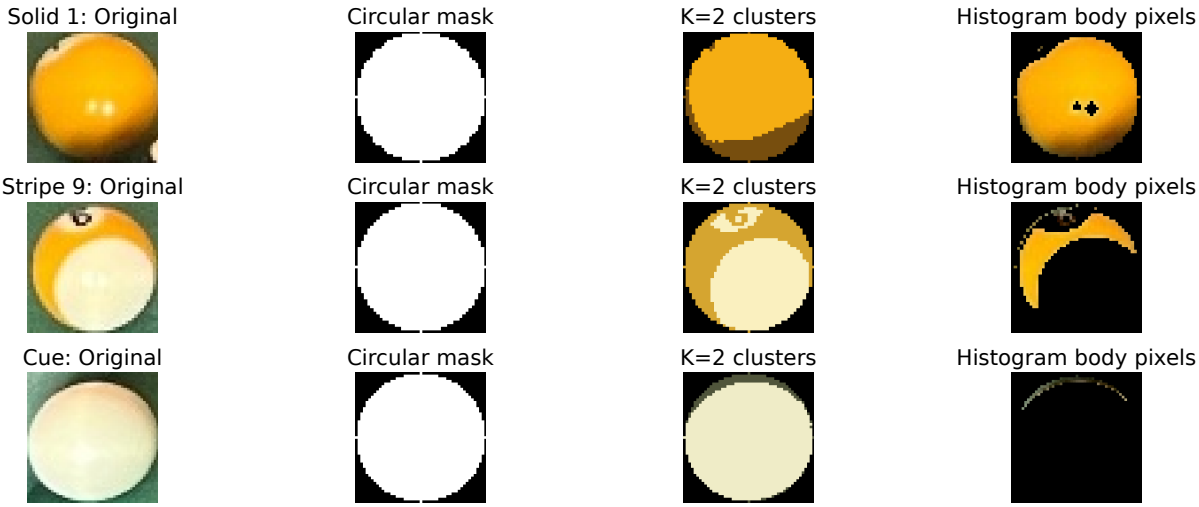


Figure 3: Original crops, extraction masks, two-cluster segmentation and selected histogram pixels. Rows show yellow solid 1, yellow stripe 9 and a cue. Black outside the masks means excluded pixels. The cue is not a body-colour reference.

Eight HSV reference histograms describe yellow, blue, red, purple, orange, green, maroon and black. AllBalls detections are matched to its labelled centres to obtain reference identities and crop radii; the 15 numbered balls supply the examples. Solid/striped partners contribute to the same colour model. The cue is excluded. A 3×3 elliptical erosion reduces crop-edge contamination; bright, low-saturation pixels are excluded using the light thresholds above. Fewer than 12 retained pixels means no numbered-colour evidence.

Histograms have $18 \times 8 \times 8 = 1152$ bins over hue $[0, 180)$ and saturation/value $[0, 256)$. Near-neutral pixels with BGR channel range below 15 receive hue zero, reducing instability in almost grey colours. A separable $[1, 2, 1]/4$ filter smooths bins, wrapping hue and clamping the other endpoints. Each descriptor sums to one; normalised reference examples are pooled and normalised again. This represents colour distributions rather than a single dominant hue, at the cost of discarding spatial arrangement. The settings are a practical compromise, not demonstrated optima.

1.3 Soft evidence and joint identity assignment

For normalised query histogram p and reference q_c, body-colour similarity is one minus Bhattacharyya distance:

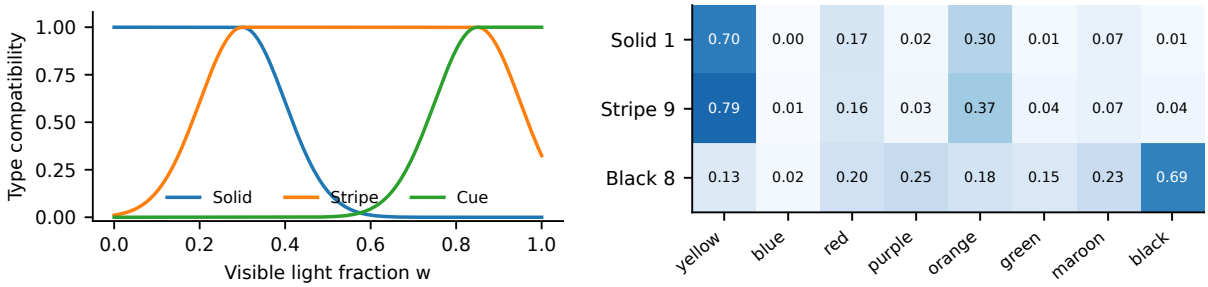
$$s_c = 1 - \sqrt{1 - \sum_b \sqrt{p_b q_{c,b}}}.$$

These eight similarities are not probabilities and need not sum to one. Colour alone cannot distinguish solid/striped partners, which share a reference.

The final recogniser retains type uncertainty. Preferred light-fraction intervals I_t are solid $[0, 0.30]$, stripe $[0.30, 0.85]$ and cue $[0.85, 1]$. With distance d to an interval (zero inside), compatibility and identity scores are

$$g_t(w) = \exp\left[-\frac{d(w, I_t)^2}{2(0.10)^2}\right], \quad S_n = s_{c(n)} g_{t(n)}(w), \quad S_0 = w g_{\text{cue}}(w).$$

Support decreases smoothly outside each interval; width 0.10 controls tolerance.



Soft type support

Measured Table2 similarities

Figure 4: Yellow partners have similar colour preferences but different white coverage.

Maximum-weight one-to-one assignment enforces each identity at most once [2]. Separate dummy choices allow multiple Unknowns. Unusable evidence or scores at most 10^{-9} are ineligible. This is not calibrated rejection: weak extra candidates can still take real identities.

Initially white ROIs are expanded by one radius pixel and reanalysed, without changing stored detection geometry. When multiple original candidates are white, the largest original Hough radius reserves cue identity. The original classifications determine this selection even if expansion changes the evidence.



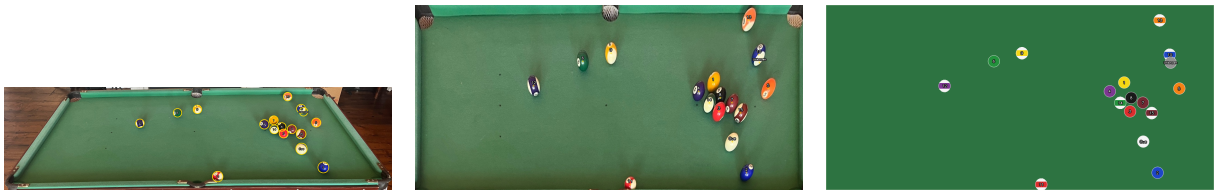
Figure 5: Table1 cue and white patch before/after expansion. The patch's light coverage falls from 100% to 93.98%, but it remains Unknown: padding is not a general repair.

1.4 Table coordinates and final representation

A homography maps the supplied corners to $(0, 0)$, $(1433, 0)$, $(1433, 682)$ and $(0, 682)$ in a 1434×683 rectangle:

$$\lambda(x_t, y_t, 1)^T = H(x, y, 1)^T.$$

Extraction uses original-image circles; transformed centres supply evaluation and overhead positions. At the supplied scale, table units are approximately millimetres. Diagram balls have nominal 43 mm diameter; their bands depict identity, not observed orientation. Because visible ball centres lie above the cloth, the cloth-plane homography can introduce systematic position error.



Original labels Rectified labels Identity diagram

Figure 6: Table2: all genuine identities are correct; an extra detection remains Unknown.

1.5 Alternatives Considered

Spatial type information. CCA measures light/non-light component counts, largest areas and light coverage in an inner boundary band. Eight-connected components smaller than 2% of ROI area are ignored in component statistics. A 5×5 erosion defines the band. Fixed rules test whether spatial features distinguish stripe whites from highlights/number patches, using identical detections and K=2 segmentations. The stripe rule needs a light component covering at least 20% of the ROI, light on at least 20% of the boundary band, and at least 10% non-light area.

Additional detections. Nearby baseline circles are grouped using disks expanded to 1.4 times their radii. Groups with at least two detections receive local Hough searches with vote threshold 22 and centre separation 12 pixels. Cloth rejection uses HSV $35 \leq H \leq 95$, $S \geq 35$, $V \geq 30$, a 9×9 table inset and 3×3 closing. Duplicate/pocket proposals are removed; baseline circles remain.

Optional watershed uses candidate centres as markers and non-foreground as background. Additional circles need basin-supported foreground area 0.15–2.0 times their disk area. Watershed filters proposals; it does not invent centres.

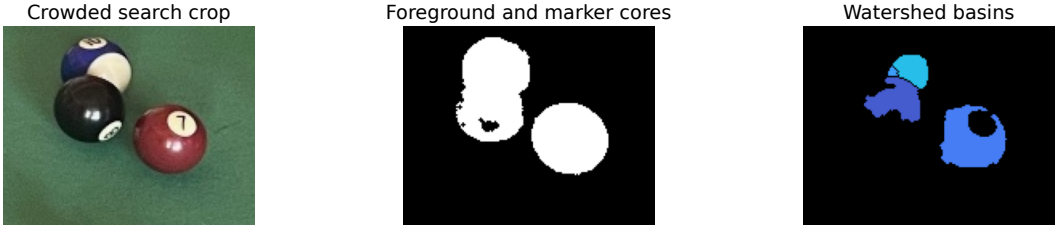


Figure 7: An actual Table3 search region, foreground with marker cores, and watershed basins. Coloured regions are segmentation labels, not ball identities.

2 Results and analysis

2.1 Evaluation protocol and final performance

Tables1–6 contain 90 annotated balls. Detected centres are matched one-to-one to annotations within 15 rectified table units. Matching maximises valid pairs; proximity orders alternatives but does not guarantee globally minimum total distance. A matched circle is a true positive (TP), an unmatched circle a false positive (FP), and an unmatched annotation a false negative (FN). Pooled counts give

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2TP}{2TP + FP + FN}, \quad \bar{e} = \frac{1}{TP} \sum_i \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2.$$

For C correctly numbered matched balls, identity accuracy is C/TP and end-to-end correct-identity recall is C/N, where N is the annotation count. Misses and Unknowns on genuine balls reduce the latter; surplus detections are penalised by detection precision, not directly by C/N. Colour evaluation excludes the six cues. AllBalls is excluded from pooled validation results because it supplies reference colours and detection tuning. The annotation file is BallLocations.csv.

Table 1: Final pipeline on every validation image. Centre error is in table units; identity fractions use matched detections.

Image	GT	TP	FP	FN	Mean error	Correct IDs
Table1	16	14	1	2	7.68	11/14
Table2	16	16	1	0	3.58	16/16
Table3	16	15	0	1	3.12	15/15
Table4	15	14	0	1	3.89	14/14
Table5	14	14	0	0	5.83	13/14
Table6	13	13	0	0	5.76	11/13
Pooled	90	86	2	4	4.91	80/86

Detection precision is **97.73%**, recall **95.56%** and F1 **96.63%**. Final identity accuracy is **93.02%** on matched balls; end-to-end recall is **88.89%**. The pooled localisation error weights each image by its matched detections, rather than averaging image means equally.

Table 2: Standalone appearance estimates versus final joint recognition.

Stage	Correct / matched	Correct / annotated
Initial type	76/86 (88.37%)	76/90 (84.44%)
Body colour, excluding cue	73/80 (91.25%)	73/84 (86.90%)
Final identity	80/86 (93.02%)	80/90 (88.89%)

These measure different stages: final identities also use soft compatibility, white-crop expansion and global uniqueness. Their gains cannot be attributed to any individual

addition without an isolated ablation. The six cues are all correct; AllBalls also gives 16/16 identities, but that is reference-image performance.

2.2 Scene-level successes and failures



Figure 8: All six validation outputs. Fractions count correct matched identities; they are not complete-scene accuracy. False detections and Unknowns remain visible.

Table2 identifies every genuine ball but retains a false circle as Unknown. Tables3 and 4 identify every match but miss blue stripe 10. Tables5 and 6 miss no balls yet make identity errors: detection and recognition fail separately.

Table1 misses balls 6 and 7, leaves stripe 14 Unknown, exchanges identities 5 and 9, and labels a false circle 6. The exchanged matches have centre errors 14.63 and 14.81, near the 15-unit tolerance. Displaced crops can corrupt recognition even when centres count as detected.

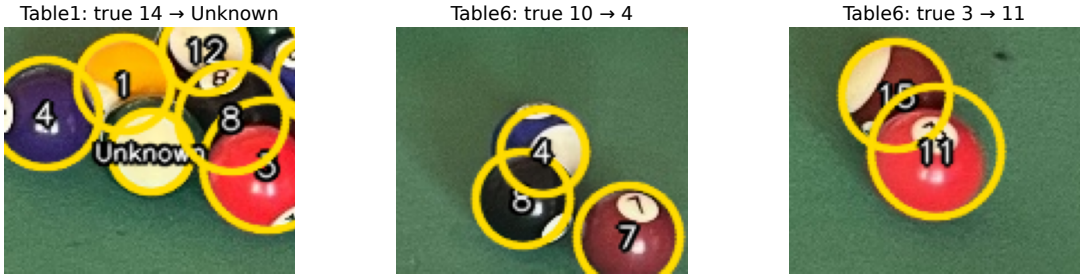


Figure 9: Table1 white patch; Table6 blue stripe 10 assigned 4; Table6 red solid 3 assigned 11. Titles give matched truth; image labels are predictions.

In Table6, red solid 3 has strongest red similarity but 66.53% estimated light coverage, favouring stripe 11. Blue stripe 10 scores 0.52 against black, 0.43 purple and 0.24 blue. Final identity 4 reflects ambiguous colour *and* unique-assignment competition, not simply the best histogram match. Visible stripe area, illumination and crop displacement are plausible contributors, not isolated causal findings. The Table1 patch still lacks usable body-colour evidence after expansion.

2.3 What the controlled experiments show

K=2 with CCA. On 86 matched balls, correct types fall from 76 to 73: zero corrections and three regressions. Two cues become striped (Tables2 and 3), as does a solid in Table4. Light-region size and boundary coverage satisfy the tested stripe rule despite these identities. Connectivity is not a unique stripe signature.

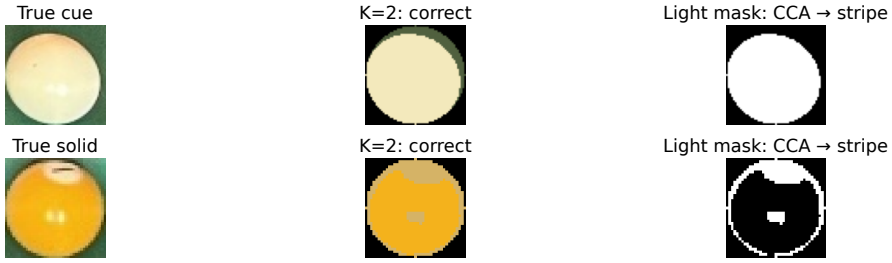


Figure 10: CCA regressions: Table2 cue and Table4 solid, both initially correct. Their light masks trigger the fixed stripe rule.

Guided detection and unchanged recognition. All variants use identical reference histograms, recognition settings and unique assignment.

Table 3: Pooled results. NFP: false detections assigned real identities. Correct identities are out of 90 annotations.

Detection	TP	FP	FN	F1 (%)	Correct IDs	NFP
Baseline	86	2	4	96.63	80	1
Guided Hough	89	8	1	95.19	81	6
Guided + watershed	88	6	2	95.65	80	5

Guided Hough recovers three misses, raising recall to 98.89%, but adds six false detections and only one net correct identity. Watershed removes two false positives and one recovered ball, leaving no identity gain. On Table3, an extra candidate takes identity 12 from a previously correct ball: assignment cannot absorb all extras.

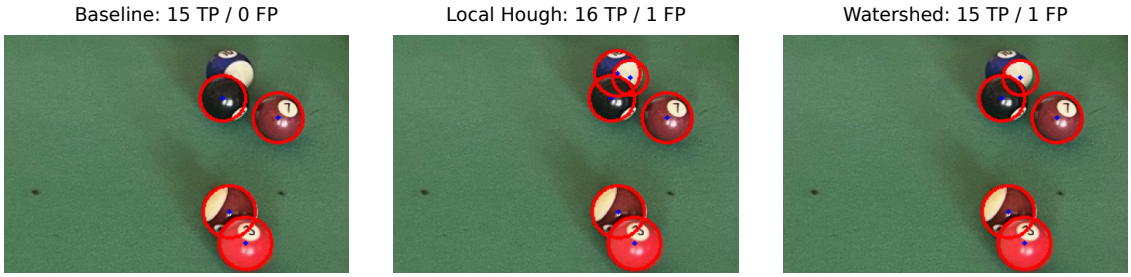


Figure 11: The same Table3 window: local Hough recovers a ball with a false circle; the watershed area check rejects the recovery but retains that false circle.

The baseline was retained for fewer false numbered outputs and lower complexity. Guided Hough remains promising for recall. These results concern the tested settings; different segmentation, markers, features or tuning could change either experiment.

2.4 Group performance, robustness and improvements

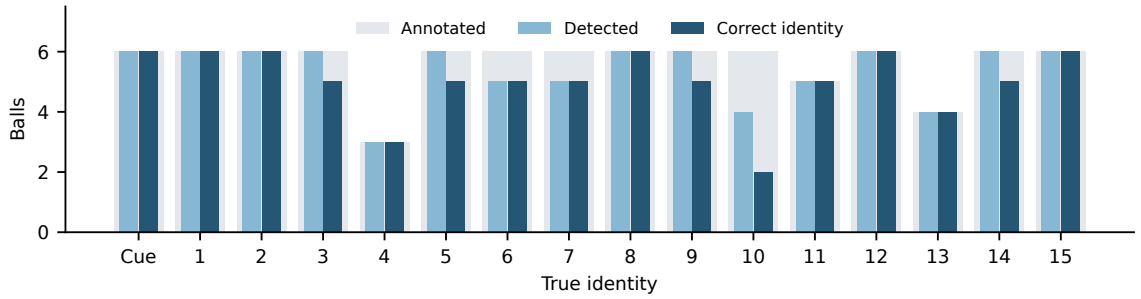


Figure 12: Counts by true identity: annotated examples, matched detections and correct final identities. Unequal sample counts remain visible.

Blue stripe 10 is weakest: six annotations yield four detections and two correct identities (33.3% end-to-end). Solids give 41/45 correct (91.1%), stripes 33/39 (84.6%) and cues 6/6. Some perfect identities have only three or four examples; these identify development weaknesses, not reliable generalisation rates.

Appropriateness and scene changes. Circular voting suits visible outlines, histograms summarise colour variation, and uniqueness uses a physical constraint. However, radius limits, supplied corners and pocket locations depend on the camera. New resolution/viewpoint requires updated geometry and size ranges. A nearer white patch can exceed a distant cue's radius. Lighting and camera response change both histograms and white coverage; occlusion removes circle and stripe evidence.

Limits of validation. Repeated inspection of six images from one scene introduces selection bias. One recovered identity changes end-to-end recall by 1.11 percentage points, insufficient to establish robustness. Freeze settings before testing new photographs with varied lighting, viewpoint, orientation and crowding. The 15-unit tolerance is about 35% of nominal ball diameter; evaluate sensitivity to stricter tolerances rather than adjusting it to raise scores. Centre annotations cannot establish segmentation or boundary accuracy.

Targeted improvements. More varied labelled references and illumination normalisation could improve colour stability. Adaptive table geometry and size priors could support new cameras. A validated Unknown policy could prevent weak proposals taking identities. Improve foreground/marker construction while tracking corrections *and* regressions: green-cloth rejection can remove green balls, and crowded windows cannot recover isolated misses. Assess detection and final identities together; attractive segmentation alone is not evidence of improved performance.

References and assistance [1] K. Dawson-Howe, CSU44053 Computer Vision course material, Trinity College Dublin.

[2] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, 2, 83–97 (1955).

ChatGPT/Codex assisted with programming, diagnostics and report drafting. Results and illustrations come from application outputs and recorded report diagnostics.